

Niklas Schweiger

M.Sc. Student & Research Assistant · TU Munich

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Master's student at TU Munich and research assistant in Prof. Daniel Cremers' Computer Vision & AI group. Work focuses on inference-time alignment of diffusion and flow models. First-author paper accepted to ECCV 2026.

EDUCATION	M.Sc. Robotics, Cognition, Intelligence Oct 2023 – Present <i>Technical University of Munich (TUM), Germany</i> • Current GPA: 1.6 (German scale) • Focus: Machine learning, generative models, inference-time optimization • Exchange: Chalmers University of Technology, Gothenburg, Sweden (Sep 2024 – Jan 2025)
	B.Sc. Electrical Engineering & Information Technology Oct 2020 – Sep 2023 <i>Technical University of Munich (TUM), Germany</i> • Final GPA: 2.5 (German scale) • Specialization: Artificial Intelligence and Machine Learning
EXPERIENCE	Research Assistant (HiWi) Mar 2026 – Present <i>Chair for Computer Vision & Artificial Intelligence (CVAI), TU Munich</i> • Developed Trust-Region Noise Search (TRS), a black-box method to align diffusion and flow models without fine-tuning or gradients; evaluated on image, molecule, and protein design. • Led the full implementation, experiments, and benchmarking against multiple baselines, as well as the writing of the paper.
	Internship – AI in Industrial Production 2023 <i>Siemens AG</i> Developed a 3D feature matching system using CNN embeddings to automate part retrieval in manufacturing databases, bridging deep learning with industrial applications.
PUBLICATIONS	Schweiger, N., Ram, K., & Cremers, D. (2026). Trust-Region Noise Search for Black-Box Alignment of Diffusion and Flow Models. <i>European Conference on Computer Vision (ECCV), 2026. Also presented at ReALM-GEN Workshop @ ICLR 2026.</i>
AWARDS & ACHIEVEMENTS	2nd Place – TUM.ai Makeathon Apr 2023 Built <i>Caire</i>, an app designed to overcome language barriers between nurses and patients using AI to extract medical information from natural speech. Awarded a €2,000 prize.
SKILLS & INTERESTS	Programming Python (Advanced), PyTorch (Advanced), C/C++ (Intermediate), Matlab (Intermediate), LaTeX
	Languages German (Native), English (C1 – Professional working proficiency)
	Interests Football, Strength training & running, Guitar (10+ years), AI & Natural sciences